

Curriculum Vitae

Rodrigo VEIGA

Full name: Rodrigo Soares Veiga.

Nationality: Brazilian.

Date of birth: May 2, 1988.

Professional address:

University of Nottingham;
School of Mathematical Sciences;
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Google Scholar: [SIzY0mgAAAAJ&hl](#)

Education

- **PhD, Physics.**
University of São Paulo; São Paulo; Brazil.
Thesis: *Statistical Physics Analysis of Machine Learning Models*.
Date of Graduation: August 4, 2022.
- **Master of Sciences, Physics.**
University of São Paulo; São Carlos; Brazil.
Thesis: *Effects of Correlated Hybridization in the Single-impurity Anderson Model*.
Date of Graduation: May 31, 2012.
- **Bachelor in Physics.**
University of São Paulo; São Carlos; Brazil.
Date of Graduation: December 21, 2009.

Research/Employment History

- **Assistant Professor.**
University of Nottingham; United Kingdom.
School of Mathematical Sciences.
Dates: September, 2025 – (current job).

- **Postdoctoral researcher.**
École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
SMILS - Lab for Statistical Mechanics of Inference in Large Systems.
Supervisor: Prof. Nicolas Macris.
Dates: October, 2022 – August, 2025.
- **Visiting PhD student.**
École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
IdePHICS - Information, Learning and Physics Lab.
Supervisor: Prof. Florent Krzakala.
Dates: February, 2021 – January, 2022.
- **PhD student.**
University of São Paulo (USP); São Paulo; Brazil.
IFUSP - Institute of Physics.
Supervisor: Prof. Renato Vicente.
Dates: July, 2017 – August, 2022.
- **Financial administrator.**
Primos Materiais para Construções Ltda; Sorocaba; Brazil.
Non-academic employment.
Dates: July, 2013 – March, 2017.
- **PhD student.**
University of São Paulo (USP); São Carlos; Brazil.
IFSC - São Carlos Institute of Physics.
Supervisor: Prof. Miled Moussa.
Dates: August, 2012 – May, 2013 (interrupted).
- **Master student.**
University of São Paulo (USP); São Carlos; Brazil.
IFSC - São Carlos Institute of Physics.
Supervisor: Prof. Valter Líbero.
Dates: March, 2010 – May, 2012.
- **Undergraduate student project.**
University of São Paulo (USP); São Carlos; Brazil.
IFSC - São Carlos Institute of Physics.
Supervisor: Prof. Valter Líbero.
Dates: April, 2008 – December, 2009.

Publications

- **Denoising Score Matching with Random Features: Insights on Diffusion Models from Precise Learning Curves.**
Authors: A. J. George, **R. Veiga**, N. Macris.
- Published: *Preprint* (2025). Under review.
- arXiv: [2502.00336](https://arxiv.org/abs/2502.00336)

- **Analysis of Diffusion Models for Manifold Data.**
Authors: A. J. George, **R. Veiga**, N. Macris.
- Published: *Preprint* (2025). Accepted *IEEE ISIT 2025*.
- arXiv: [2502.04339](https://arxiv.org/abs/2502.04339)
- **Stochastic gradient flow dynamics of test risk and its exact solution for weak features.**
Authors: **R. Veiga**, A. Remizova, N. Macris.
- Published: *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR [235:49310-49344](https://proceedings.mlr.press/v235/veiga24a.html) (2024).
- arXiv: [2402.07626](https://arxiv.org/abs/2402.07626)
- **Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks.**
Authors: **R. Veiga**, L. Stephan, B. Loureiro, F. Krzakala, L. Zdeborová.
- Published: *Advances in Neural Information Processing Systems (NeurIPS)*. Volume 35; [pages 23244-23255](https://proceedings.neurips.cc/paper_files/paper/2022/file/veiga22a.pdf) (2022).
- Re-published: *Journal of Statistical Mechanics; Theory and Experiment*. Volume 2023, page 114008, DOI [10.1088/1742-5468/ad01b1](https://doi.org/10.1088/1742-5468/ad01b1) (2023).
- arXiv: [2202.00293](https://arxiv.org/abs/2202.00293)
- **Learning curves for the multi-class teacher–student perceptron.**
Authors: E. Cornacchia*, F. Mignacco*, **R. Veiga***, C. Gerbelot, B. Loureiro, L. Zdeborová (*equal contribution).
- Published: *Machine Learning: Science and Technology*. Volume 4; pages 05019; DOI [10.1088/2632-2153/acb428](https://doi.org/10.1088/2632-2153/acb428) (2023).
- arXiv: [2203.12094](https://arxiv.org/abs/2203.12094)
- **Restricted Boltzmann machine flows and the critical temperature of Ising models.**
Authors: **R. Veiga**, R. Vicente.
- Published: *Preprint* (2020).
- arXiv: [2006.10176](https://arxiv.org/abs/2006.10176)
- **Age-structured estimation of COVID-19 ICU demand from low quality data.**
Authors: **R. Veiga**, R. Murta, R. Vicente.
- Published: *Preprint* (2020).
- arXiv: [2006.06530](https://arxiv.org/abs/2006.06530)

Participation in events

Oral presentations

- **The Abdus Salam International Centre for Theoretical Physics (ICTP); Trieste, Italy.**
Contributed talk: *Time Evolution of the Test Risk under Stochastic Gradient*

Flow Dynamics.

Event: Youth in High Dimensions: Recent Progress in Machine Learning, High-Dimensional Statistics and Inference.

Date: May, 2024.

- **African Institute for Mathematical Sciences (AIMS); Kigali, Rwanda.**
Invited speaker: *Time Evolution of the Test Risk under Stochastic Gradient Flow Dynamics.*
Event: From Theory to Practice: Workshop in Data Science.
Date: April, 2024.
- **TOPML, Workshop on the Theory of Overparameterised Machine Learning; Houston, USA.**
Contributed talk: *Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks.*
Virtual event.
Date: April, 2022.

Poster presentations

- **ICML, International Conference on Machine Learning; Vienna; Austria.**
Poster title: *Stochastic Gradient Flow Dynamics of Test Risk and its Exact Solution for Weak Features.*
Date: July, 2024.
- **Institut d'Études Scientifiques de Cargèse; Cargèse, France.**
Poster title: *Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks.*
Event: Statistical Physics and Machine Learning Back Together.
Date: August, 2023.
- **NeurIPS, Conference on Neural Information Processing Systems; New Orleans, USA.**
Poster title: *Phase diagram of stochastic gradient descent in high-dimensional two-layer neural networks.*
Date: December, 2022.
- **University of São Paulo, São Carlos Institute of Physics; São Carlos, Brazil.**
Poster title: *Entanglement and quantum Discord in the superradiance.*
Event: II SIFSC - São Carlos Physics Institute Graduate Workshop.
Date: October, 2012.
- **University of São Paulo, São Carlos Institute of Physics; São Carlos, Brazil.**
Poster title: *Effects of correlated hybridization in the single-impurity Anderson model.*
Event: I SIFSC - São Carlos Physics Institute Graduate Workshop.
Date: October, 2011.

- **Federal University of Rio Grande do Norte, IIP; Natal, Brazil.**
Poster title: *Effects of correlated hybridization in the single-impurity Anderson model.*
Event: Brazilian School on Statistical Mechanics.
Date: July, 2011.
- **University of São Paulo, São Carlos Institute of Physics; São Carlos, Brazil.**
Poster title: *Correlated hybridization in the single-impurity Anderson model and non-local functional in the Heisenberg model.*
Event: XIV São Carlos Physics Institute Graduate Workshop.
Date: October, 2010.

Schools

- **ICTP South American Institute for Fundamental Research; São Paulo, Brazil.**
 Event: First School on Data Science and Machine Learning.
 Date: December, 2019.
- **ICTP South American Institute for Fundamental Research; São Paulo, Brazil.**
 Event: Minicourse on Machine Learning for Many-Body Physics.
 Date: September, 2017.

Scholarships

- **CAPES-PrInt, Program for Institutional Internationalization; Brazil.**
 Scholarship for a PhD internship abroad.
Project: Statistical physics inference on machine learning algorithms.
Grant number: 88887.467036/2019-00.
Host: École Polytechnique Fédérale de Lausanne; Switzerland.
Dates: February, 2021 – January, 2022.
- **CNPq, The National Council for Scientific and Technological Development; Brazil.**
 PhD scholarship.
Project: Statistical physics and machine learning models.
Grant number: 162857/2017-9.
Dates: August, 2017 – February, 2020.
- **FAPESP, The State of São Paulo Research Foundation; Brazil.**
 PhD scholarship.
Project: Entanglement and quantum discord in the superradiance and applications of quantum information theory in NMR.
Grant number: 2012/12065-7.
Dates: Declined.
- **CAPES, Coordination for the Improvement of Higher Education Personnel; Brazil.**
 PhD scholarship.
Project: Entanglement and quantum discord in the superradiance and

applications of quantum information theory in NMR.

Grant number: PROEX.

Dates: August, 2012 – May, 2013.

- **FAPESP, The State of São Paulo Research Foundation; Brazil.**

Master scholarship.

Project: Effects of correlated hybridization in the single-impurity Anderson model.

Grant number: 2009/13065-8.

Dates: March, 2010 – February, 2012.

- **FAPESP, The State of São Paulo Research Foundation; Brazil.**

Undergraduate research scholarship.

Project: *Density functional theory applied to the antiferromagnetic Heisenberg model.*

Grant number: 2007/59988-4.

Dates: April, 2008 – December, 2009.

Teaching

- **CS526 Learning theory, EPFL; Lausanne; Switzerland.**

Master's course whose main instructor is Prof. Nicolas Macris.

Spring semester 2024

Responsible for two lectures:

- March 18, 2024: Bias variance tradeoff and the double descent phenomenon.
- March 25, 2024: Double descent and the weak features model.

Spring semester 2025

- February 17, 2025: PAC learning framework and uniform convergence.
- March 17, 2025: Bias variance tradeoff and the double descent phenomenon.
- March 24, 2025: Double descent and the weak features model.

Academic service

- **Reviewer**

- Physical Review E.
- Physical Review Letters.
- International Conference on Machine Learning (ICML).
- Conference on Neural Information Processing Systems (NeurIPS).
- IEEE International Symposium on Information Theory (ISIT).

- **Member of the Master's Dissertation Committee**

Student: Pedro Henrique Mendes Duarte

Title: *Applications of Restricted Boltzmann Machines in Lattice Models*

Institute of Physics, Federal University of Rio Grande do Sul, Brazil.

Date: January 24, 2025.

Outreach project

- **Modeling growth rate and growth acceleration rate of COVID-19 cases in Brazil.**
Code available on [GitHub](#).
Brazilian newspaper *Folha de S. Paulo* used this model to construct an [acceleration monitor](#) of COVID-19 in Brazil.
 - Several press reports followed from the acceleration monitor based on the project: [[1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#), [10](#)]

References

- **Prof. Nicolas Macris.**
École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
IC - School of Computer and Communication Sciences.
SMILS - Lab for Statistical Mechanics of Inference in Large Systems.
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- **Prof. Florent Krzakala.**
École Polytechnique Fédérale de Lausanne (EPFL); Lausanne; Switzerland.
STI - School of Engineering.
IdePHICS - Information, Learning and Physics Lab.
florent.krzakala@epfl.ch
- **Dr. Bruno Loureiro.**
École Normale Supérieure - PSL & CNRS; Paris; France.
Département d'Informatique.
Center for Data Science.
bruno.loureiro@di.ens.fr
- **Prof. Renato Vicente.**
University of São Paulo (USP); São Paulo; Brazil.
IME - Institute of Mathematics and Statistics.
MAP - Applied Mathematics Department.
rvicente@usp.br
- **Prof. Nestor Caticha.**
University of São Paulo (USP); São Paulo; Brazil.
IF - Institute of Physics.
FGE - General Physics Department.
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